A Biomimetic Solar Energy-Enhanced Outdoors Research Platform Enabling Infinite UAV Endurance and Multi-Terrain Flexibility

Abstract - Unmanned aerial vehicles (UAVs) are widely applied in various scenarios—including aerial photography, agricultural spraying, environmental monitoring, and disaster response—but their operational capabilities are severely constrained by limited battery life and the need for frequent recharging. Current UAV technologies primarily rely on battery power and fixed charging stations, restricting both range and efficiency. This limitation is particularly problematic in ecologically sensitive regions such as Santa Barbara, California, an important habitat and migratory corridor for birds, where traditional observation methods (e.g., manual tracking and fixed cameras) can disturb wildlife and offer limited coverage. To overcome these challenges, This paper proposes a “self-sustaining” UAV system designed to achieve theoretically infinite flight time by autonomously harvesting renewable energy. This approach integrates several essential functions: photovoltaic panels to capture solar energy; a dual-axis, threshold-triggered sun-tracking system to maximize energy conversion efficiency; and adaptive biomimetic attachment mechanisms—featuring suction cups and grippers—that allow the UAV to securely adhere to diverse surfaces for charging and monitoring. A flotation device is also incorporated to enable safe landings on uneven terrain or water, enhancing operational versatility. By leveraging solar energy and adaptive anchoring mechanisms, the proposed platform eliminates the need to return to fixed charging stations, enabling longer missions with higher efficiency. This design offers a sustainable solution that can revolutionize UAV technology, paving the way for extended field operations in wildlife monitoring, environmental research, and disaster response, among other applications.

Keywords: UAV, biomimetics, solar panel, endurance, suction cup, gripper, dual-axis sun-tracking, renewable energy, perpetual flight

1. Introduction

1.1 Research Background and Significance

Due to its unique geographical location and diverse ecological environment, California's Santa Barbara region has become a crucial bird migration and habitat area. Research on the birds in

this region holds significant ecological value, especially for understanding migration patterns, population dynamics, and ecological adaptability. Local researchers have long focused on studying seabird breeding behavior, migration routes, and the impacts of climate change on their survival. Such studies help reveal how birds adapt and persist amid global climate change and habitat alterations, providing invaluable data for bird conservation and ecological restoration efforts worldwide.

Traditional observation methods, such as manual monitoring and fixed-camera systems, face considerable limitations. Manual observations require extensive human resources and cannot cover vast ecological areas, while fixed cameras, though capable of continuous recording, offer limited observation ranges and lack the flexibility to adapt to rapidly changing environmental conditions and bird migration behaviors. Consequently, after careful consideration, a more efficient and non-intrusive technological approach has been pursued for this project.

Despite the transformative impact of UAV-based bird observation on wildlife research, a major challenge remains: the limited endurance of UAVs. Traditional approaches to extending flight times—such as increasing battery capacity, using more efficient motors and propulsion systems, or establishing fixed charging stations—come with drawbacks. Increasing battery capacity often results in added weight, adversely affecting flight performance. At the same time, fixed charging stations are costly and inflexible, rendering them unsuitable for the varied research demands and extensive monitoring areas encountered in the field.

Additionally, some studies have attempted long-term monitoring by attaching sensors to animals. Although these sensors can collect a wealth of physiological and behavioral data, they often interfere with the natural behavior of the animals, particularly if the sensors are too heavy or poorly designed, which can alter the animals' normal activities or even influence their habitat preferences. Therefore, achieving long-term, effective monitoring without disturbing the natural behavior of wildlife remains an urgent challenge.

It is essential to develop a UAV system capable of autonomously harvesting energy and achieving long-duration flight to overcome the limitations of traditional observation methods and current UAV technology. Such a system would not only address the battery endurance issue but also minimize disturbances to bird behavior, enabling truly non-intrusive monitoring. Moreover, this innovative UAV platform can adapt to complex environments and varied research needs, significantly enhancing the efficiency and accuracy of avian ecological studies.

The development and application of this advanced UAV system promise to enable deeper and broader bird observations, ultimately providing a more solid scientific basis for global ecological and species conservation. Therefore, research into autonomous energy technologies for UAVs holds significant theoretical importance and vast practical application potential—particularly in wildlife monitoring, ecological research, and environmental protection, where it demonstrates enormous promise for future development.

1.2 Literature Review

1.2.1 Lightweight Design

Lightweight design has long been recognized as a fundamental strategy to improve UAV endurance. By reducing the weight of the UAV, researchers can enhance energy efficiency and extend flight times. Various materials and manufacturing techniques have been explored to achieve lightweight UAV designs, with carbon fiber composites and expanded polystyrene (EPS) foam emerging as key materials. These materials offer an excellent balance of strength and weight, allowing UAVs to carry more payload while minimizing energy consumption. Additive manufacturing (3D printing) has also been employed to optimize design, making it possible to produce lighter, more efficient components.[1][2]

1.2.2 Wireless Charging System

Researchers have also focused on developing efficient charging station systems to optimize layout and improve charging efficiency, meeting the diverse needs of UAV operations. For instance, Ali Bin Junaid et al. (2016) proposed a wireless charging technology combined with a fixed charging station solution, enabling UAVs to automatically charge after completing missions.[3] However, such solutions are costly, and the charging speed and efficiency may affect UAV mission execution. Moreover, fixed charging stations in field environments face high construction and maintenance costs. Ali et al. (2022) designed a novel solar-powered mobile charging station for UAVs, which includes modules such as maximum power point trackers, constant current/voltage chargers, and battery selection systems.[4] However, since the charging station and UAV are separate entities, issues related to the return time and energy consumption for recharging still persist.

1.2.3 Solar-Power/Fuel Cells For UAVs

Researchers have proposed various innovative solutions to tackle the UAV endurance issue. First, one key research direction is using photovoltaic materials to convert solar energy into electrical energy, thereby enabling prolonged flight. Studies by Luqman et al., Lin et al. (2020), and Goh et al. (2019) have systematically reviewed and demonstrated that solar-powered UAVs

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[1] Sharun, S. M., Iberahim, R., Kasim, M. S., Yusof, S. N. A., Halim Yap, M. F. A. A., & Ab Aziz, A. A. (2024). Development of a Cost-Effective UAV for Extended Duration Disaster Management Operations using Expanded Polystyrene. J. Adv. Res. Appl. Sci, 47, 1-5.

[2] Goh, G. D., Agarwala, S., Goh, G. L., Dikshit, V., Sing, S. L., & Yeong, W. Y. (2017). Additive manufacturing in unmanned aerial vehicles (UAVs): Challenges and potential. Aerospace Science and Technology, 140 – 151. https://doi.org/10.1016/j.ast.2016.12.019

[3] Junaid, A. B., Lee, Y., & Kim, Y. (2016). Design and implementation of autonomous wireless charging station for rotary-wing UAVs. Aerospace Science and Technology, 253 – 266. https://doi.org/10.1016/j.ast.2016.04.023

[4] Ali, E., Fanni, M., & Mohamed, A. M. (2022). A new battery selection system and charging control of a movable solar-powered charging station for endless flying killing drones. Sustainability, 14, 2071. https://doi.org/10.3390/su14042071

can significantly extend flight time.[5][6] Chong Swee Goh et al. (2019) and Ching-Fuh Lin et al. (2020) have proposed that a quadrotor UAV powered by solar energy can achieve extended flight time, analyzing the impact of solar power on flight performance.[7] Mickael Cosson et al. (2022) developed an energy management simulator for UAVs that includes photovoltaic generation, energy storage units, and energy management systems.[8] Studies on photovoltaic panels have revealed that to achieve “infinite endurance,” the primary challenge is to optimize the power output of the photovoltaic panel. In addition, research has explored methods to combine the energy stored in batteries with the power generated by photovoltaic cells during flight.[9] Other scholars have suggested using fuel cell systems—such as hydrogen fuel cells and direct methanol fuel cells (DMFCs)—to provide high specific power and energy for extended UAV endurance.[10] Although solar energy and fuel cells will help extend flight duration, they will not be enough to constantly sustain the UAV before eventually needing to land, meaning that it won't achieve “infinite endurance.”

1.2.4 Solar Tracking Systems

Advanced maximum power point tracking (MPPT) techniques, improved photovoltaic materials, and intelligent solar tracking control systems can enhance overall energy conversion efficiency.[11] For example, satellites use MPPT technology—by continuously monitoring the photovoltaic system’s output voltage and current, they dynamically adjust the operating point so that the panels always operate at maximum power under varying light and temperature conditions. Although effective, such methods involve complex algorithms and high implementation difficulty. Wei Chen (2011) proposed using multiple light-sensitive components to capture sunlight from different directions for solar tracking, while Parashar and Kumar (2021) noted that solar tracking systems can be classified into single-axis, dual-axis, and inclined trackers—with dual-axis systems providing higher energy output compared to fixed installations (see Kumba et al. (2024)).[12][13][14] However, dual-axis systems require multiple adjustments that consume energy. To optimize the photovoltaic panel’s energy input versus adjustment consumption, prediction-based tracking algorithms (such as astronomical algorithms and fuzzy logic control) can significantly

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[5] Luqman, S., Al, A., Afzal, A., Khalifa, O., Abri, A., & Khan, A. (n.d.). Solar-Powered UAVs: A systematic Literature Review.

[6] Goh, C. S., Kuan, J. R., Yeo, J. H., Teo, B. S., & Danner, A. (2019). A fully solar‐powered quadcopter able to achieve controlled flight out of the ground effect. Progress in Photovoltaics: Research and Applications, 869–878. https://doi.org/10.1002/pip.3169

[7] Lin, C., Lin, T., Liao, W., Lan, H., Lin, J., Chiu, C., & Danner, A. (2020). Solar Power Can Substantially Prolong Maximum Achievable Airtime of Quadcopter Drones. Advanced Science. https://doi.org/10.1002/advs.202001497

[8] Cosson, M., David, B., Arzel, L., Poizot, P., & Rhallabi, A. (2022). Modelling of photovoltaic production and electrochemical storage in an autonomous solar drone. eScience, 2, 235-241. https://doi.org/10.1016/j.esci.2022.02.004

[9] Chu, Y., Ho, C., Lee, Y., & Li, B. (2021). Development of a Solar-Powered Unmanned Aerial Vehicle for Extended Flight Endurance. Drones, 44. https://doi.org/10.3390/drones5020044

[10] Pan, Z. F., An, L., & Wen, C. Y. (2019). Recent advances in fuel cells based propulsion systems for unmanned aerial vehicles. Applied Energy, 473–485. https://doi.org/10.1016/j.apenergy.2019.02.079

[11] Dada, M., & Popoola, P. (2023). Recent advances in solar photovoltaic materials and systems for energy storage applications: a review. Beni-Suef University Journal of Basic and Applied Sciences.

[12] Chen, W. (2011). Designing of Solar Cell Panel Sun Chasing System on Basis of MSP430 Micropower Single Chip Microcomputer. Applied Mechanics and Materials, 345–349. https://doi.org/10.4028/www.scientific.net/amm.65.345

[13] Parashar, S., & Kumar, S. (n.d.). A Review on Solar Tracking System.

[14] Kumba, K., Upender, P., Buduma, P., Sarkar, M., Simon, S. P., & Gundu, V. (2024). Solar tracking systems: Advancements, challenges, and future directions: A review. Energy Reports, 12, 3566-3583.

enhance the energy capture efficiency of photovoltaic components.[15][16][17] In recent years, energy-efficient IoT-based DAS tracking systems have also emerged as a trend.

1.3 Project Innovations

In view of both bird observation and mobility requirements, this project's design for an infinite endurance research UAV platform adopts a lightweight design. It integrates a mobile-fixed charging station concept with a combined photovoltaic and battery power supply method, ultimately aiming to achieve "infinite endurance."

1.3.1 Integration of Multiple Functional Modules

The project, taking into account the motion characteristics of UAVs and the demands of field operations, designs three static operational modes—buoyancy (floating), grasping, and suction—allowing the UAV to perform charging via onboard photovoltaic panels. This design realizes a "mobile-fixed" charging station concept and allows UAVs to safely land/attach to surfaces.

1.3.2 Threshold-Triggered Dual-Axis Solar Tracking Strategy

Based on the above considerations of performance, execution efficiency, and economic cost, this project adopts an astronomical algorithm for solar tracking with dual-axis adjustment. The energy conversion efficiency is increased by dynamically adjusting the orientation of the photovoltaic panels to maximize solar capture. By incorporating a threshold-triggered dual-axis solar tracking strategy, which sets a minimum triggering threshold for angle changes, the photovoltaic panel's orientation is only adjusted when the sun's position has changed sufficiently, thereby balancing tracking precision and energy consumption.

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2. Mechanical Design

2.1 Overall Design

2.1.1 Version 1

The platform consists of a quadrotor UAV body, two static mechanical devices (biomimetic suction cup and claw system), a flight control system, and a photovoltaic panel system

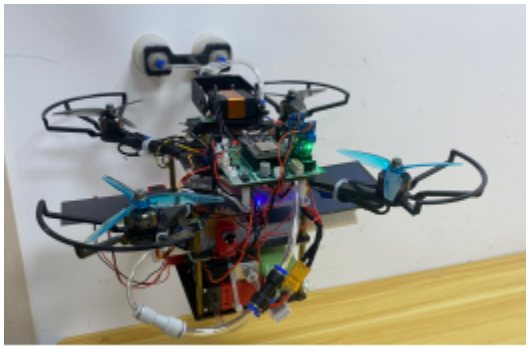
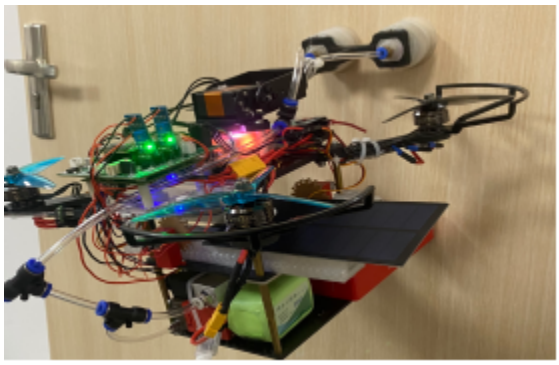
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[15] Chowdhury, M. E., Khandakar, A., Hossain, B., & Abouhasera, R. (2019). A Low‐Cost Closed‐Loop Solar Tracking System Based on the Sun Position Algorithm. Journal of Sensors, 2019(1), 3681031.

[16] Batayneh, W., Owais, A., & Nairoukh, M. (2013). An intelligent fuzzy based tracking controller for a dual-axis solar PV system. Automation in Construction, 29, 100-106.

[17] Lehloka, M., Swart, J., & Hertzog, P. (2020). A comparison of two automatic solar tracking algorithms. In E3S Web of Conferences (Vol. 152, p. 02009). EDP Sciences.

(including a sun-tracking device). It is designed to provide an integrated solution for wildlife observation missions. The design fully considers the platform's application requirements in different environments (such as forests and glass surfaces). Therefore, the design of each component not only focuses on functionality but also emphasizes mobility, load-bearing capacity, and structural strength in diverse environments.

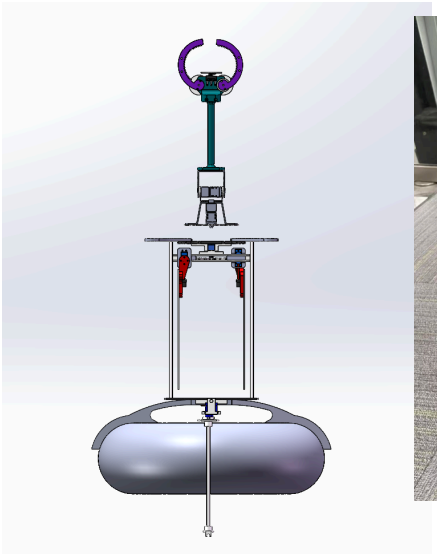
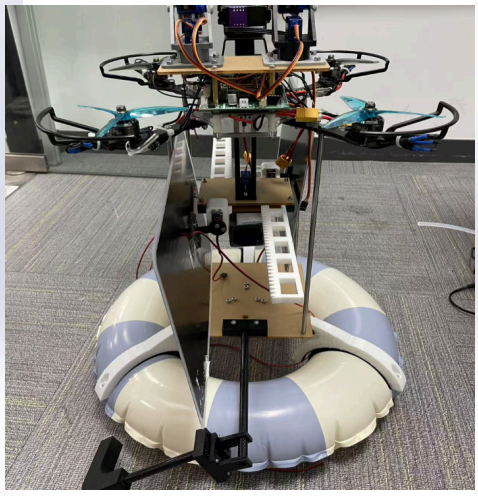
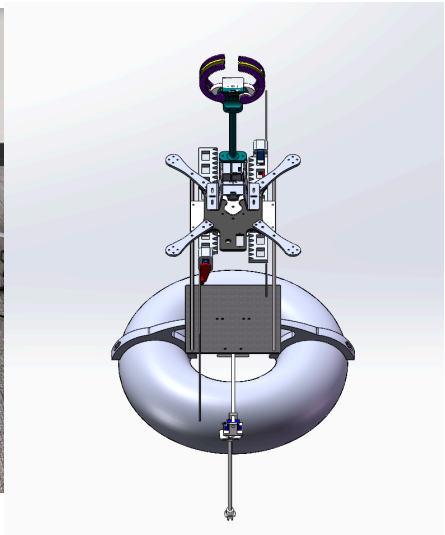



In this design, the solar panels were too small to support efficient UAV charging, and increasing size/mass led to low TWR.

2.1.2 Version 2

The platform consists of a quadcopter UAV body, three static state designs (biomimetic suction cups, claw systems, and buoyancy devices), a flight control system, and a solar panel system (including a solar tracking system). The design considers the project's need for wildlife observation in various locations such as forests, smooth surfaces (urban/suburban areas), ponds, beaches, etc. Simulation tests ensure that parts are lightweight yet sufficiently rigid for operation.

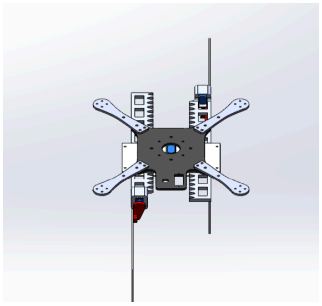
The platform's top section features interchangeable modules connected to the UAV body via an adapter, such as a mechanical claw or suction cup device. Below the UAV, a solar panel adjustment system is installed to optimize energy input during operation. A buoyancy device is attached beneath the solar panel for water surface or uneven surface operations.

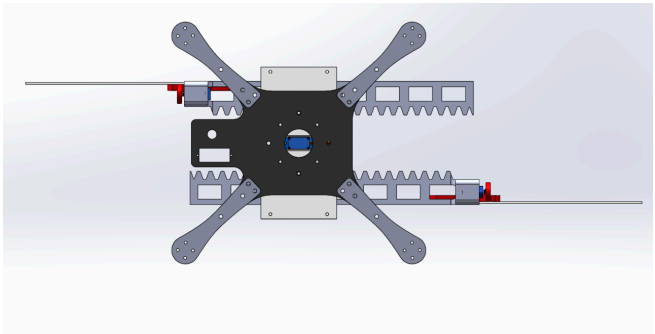




The design also integrates a modular experiment board using an ESP32 microcontroller. This board includes interfaces for suction cups, image transmission, claws, and solar panels, with serial communication for remote control.

2.2 Quadcopter UAV Body

The basis of this research platform is a quadcopter design, adhering to lightweight principles. Carbon fiber plates and 3D-printed components are used to reduce overall weight while improving flight stability and endurance. The main body consists of four brushless motors mounted with M4 screws to form the quadcopter structure.





2.3 Three Static Designs

Based on the requirements of outdoor observation, especially for bird observation tasks that involve flight-to-static coordinated missions, the UAV platform is designed with three static operation modes: grasping, suction, and floating. These correspond to three mechanical structures: a biomimetic suction cup, a gripper system, and a buoyancy device.

2.3.1 Biomimetic Suction Cup

The suction cup mechanism is one of the most common attachment devices. Typically, a single suction cup is used to adhere to flat surfaces. In 2021, Tsukagoshi et al. proposed a novel hybrid suction cup technique that comprises a hollow truncated conical support, an annular adhesive gel covering the bottom periphery of the support, and a stored release balloon within the support.[18] This design allows stable adhesion on various objects and environments. Sensen Liu et al. proposed a new dual-elastic combination suction cup and an aerial manipulation system based on it for perching or grasping tasks; their design introduced a novel self-sealing structure within the suction cup.[19] Inspired by the internal negative pressure and external

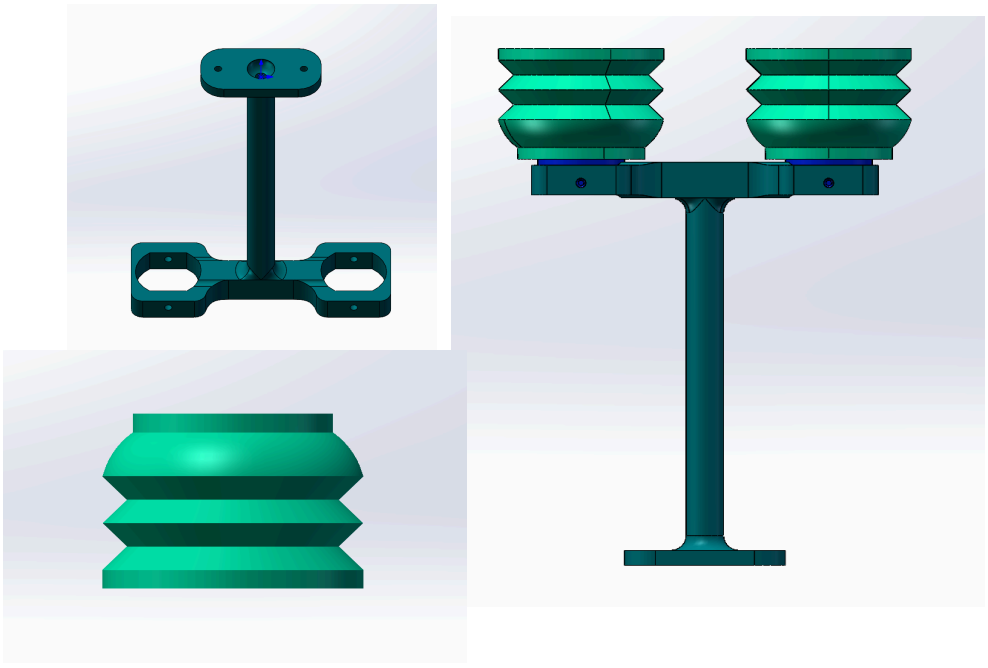
[18] Tsukagoshi, H., & Osada, Y. (2021). Soft Hybrid Suction Cup Capable of Sticking to Various Objects and Environments. Actuators, 50. https://doi.org/10.3390/act10030050

[19] Liu, S., Dong, W., Ma, Z., & Sheng, X. (2020). Adaptive Aerial Grasping and Perching With Dual Elasticity Combined Suction Cup. IEEE Robotics and Automation Letters, 4766 – 4773. https://doi.org/10.1109/lra.2020.3003879

pressure regulation of octopus suckers—which are key to their adhesion ability—the present platform adopts a dual-suction cup structure.[20] An inner and outer suction cup mechanism is used in each set to ensure firmer adhesion. Customized nylon components provide structural support for the suction cups, which are assembled as a single unit for testing. These components could not be made with basic 3D printing, as they were often porous and leaked air.

To enhance adhesion performance, the project designs two suction cup assemblies (left and right) to increase the contact area with the target surface. In addition, compression sealing is utilized at connection points between nylon and silicone parts, ensuring that the suction cups remain completely airtight while retaining a simple structure. This airtight seal is critical because it allows the system, which utilizes a 13L/min air pump, to achieve and maintain the necessary vacuum without frequent re-pumping quickly. With a pressure sensor to monitor the adhesion status in real time, the pump is triggered to apply additional pressure to the suction cup when the adhesion force drops below a critical value, effectively preventing accidental detachment.

In practical implementation, a modeled adapter is designed to effectively connect the silicone-based suction cup (chosen for its excellent elasticity and adhesion) with the UAV body. The adapter is designed to consider the geometric shape of the suction cup and the environmental requirements to ensure stable adhesion during operation. This suction device is ultimately integrated into the UAV's main structure, enabling effective perching operations on designated surfaces. This design aims to enhance the UAV's operational capability in complex environments and expand its application scope, providing theoretical and practical support for subsequent research and development.



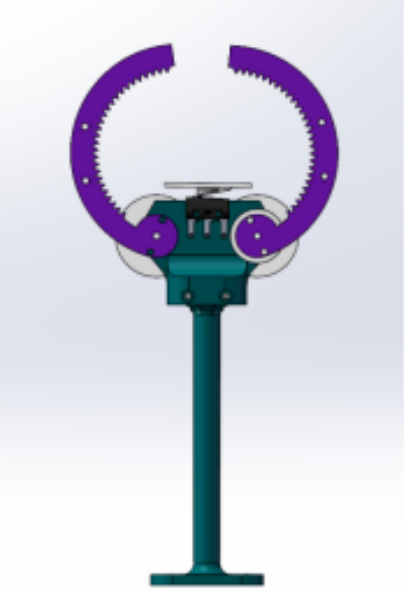
[20] Kier, W. M., & Smith, A. M. (1990). The morphology and mechanics of octopus suckers. The Biological Bulletin, 178(2), 126-136.

2.3.2 Claw System

The gripper is an important end-effector in UAV applications, particularly for tasks that require grasping or perching. In nature, various creatures exhibit gripping structures, such as birds' talons and primates' hands. Some grippers use linkage mechanisms driven by motors to open and close the gripper, providing stable grip force and adapting to objects of varying shapes and sizes.[21]

Engineered specifically for this platform, the claw is designed to mimic how bats cling to tree branches, allowing the UAV to "hang" and stabilize itself during operations. In practical use, the UAV's control system directs the claw to open and maneuver toward the target branch. Upon contact, an integrated trigger mechanism coupled with a pressure sensor—drawing inspiration from the rapid closure of a Venus flytrap—detects the branch and initiates the closing sequence. Safety thresholds, set through both button and pressure sensor feedback, ensure that the claw only opens or closes when conditions are safe, preventing inadvertent releases or damage to the target object.

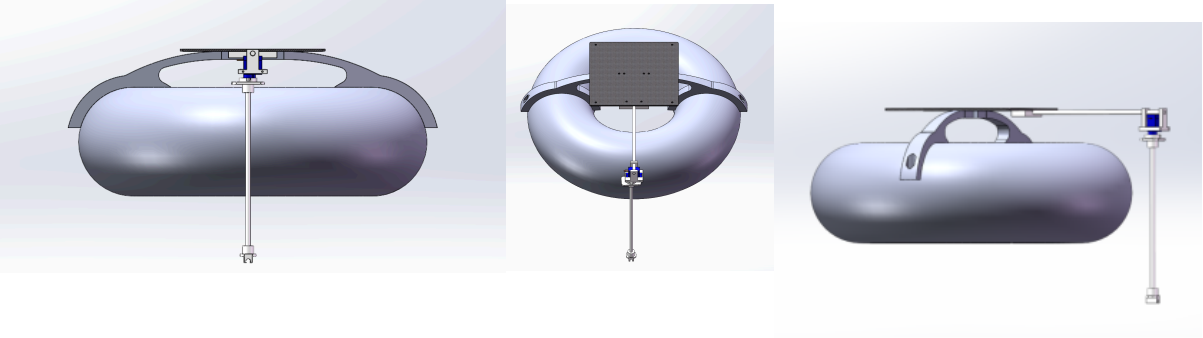
The claw features a serrated design constructed from lightweight yet rigid carbon fiber to enhance gripping performance. Structurally, it incorporates a symmetrical gear transmission system, with a central support rod connecting the base to two arc-shaped racks. This arrangement ensures synchronized movement for a consistent and reliable grip. The worm gear mechanism not only provides a strong, self-locking grip but also allows the system to maintain hold without continuous power, offering an advantage over traditional servo-based designs. This combination of biomimicry, advanced materials, and integrated safety measures makes the claw a versatile and effective tool well-suited for various challenging environments.



[21] Zhao, Y., Xiang, R., Li, H., Wang, C., Zhang, J., Liu, X., & Hao, Y. (2024). Design and Validation of a Biomimetic Leg-Claw Mechanism Capable of Perching and Grasping for Multirotor Drones. Biomimetics, 10(1), 10.

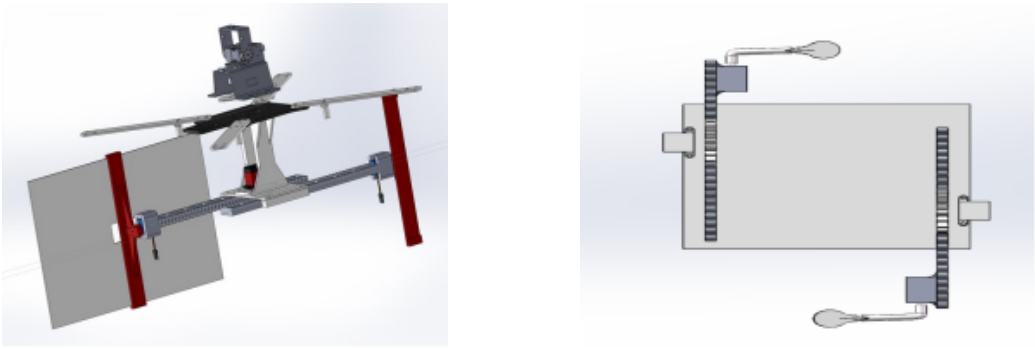
2.3.3 Buoyancy Device

The buoyancy device functions similarly to a “swimming ring.” It employs an annular buoyancy structure with a mechanical bracket to fix the upper control modules to the device. Specific mounting holes are designed on the carbon fiber plate to ensure a secure connection between the swimming ring and the UAV structure, providing stable water support. The device consists of an inflatable swimming ring that, with the aid of an inflation valve, can be rapidly inflated or deflated to increase buoyancy effectively. A rod is suspended from the bottom, ending in a motor drive mechanism, which leverages the pump to propel the UAV through water. This design allows the UAV to float on water surfaces when necessary. At the same time, it enables the UAV to perform operations on uneven land surfaces by inflating the device halfway, allowing it to conform to the terrain so the UAV can land safely and securely.



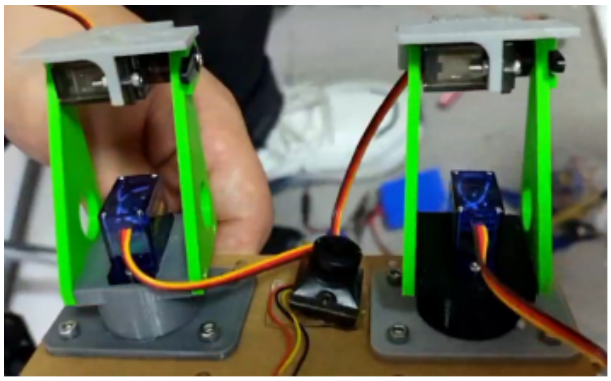
2.4 Solar Panel System

The solar panel system is designed to both securely stow the panels during flight and dynamically optimize their orientation for maximum energy capture. A robust four-arm support structure equipped with rack-and-pinion mechanisms allows the panels to extend and retract safely, protecting them during transit. Integrated into the system is a dual-axis gimbal that plays a critical role in solar tracking: the upper section—mounted between the gripper/suction cup—provides a full 360° rotation for azimuth alignment, while the lower section—controlled by a dedicated servo motor—adjusts the panel’s elevation through a 0°–180° range. This combined mechanism ensures that the solar panels continuously track the sun throughout the day, maximizing energy conversion whether the panels are deployed for charging or safely stowed during flight.



2.5 Camera Synchronization Gimbal Design

Two two-axis gimbals are installed at the rear end of the UAV for bird/wildlife observation. The gimbals are synchronized and feature one wide-angle video transmission camera and a dedicated recording camera. The recording camera relies on the video transmission camera for positioning while carrying out the recording task.



The video transmission camera will also be guiding actions such as claw actuation, suction cup actuation, and general flight.

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3. Electronics

3.1 UAV Flight Controller Circuit Design

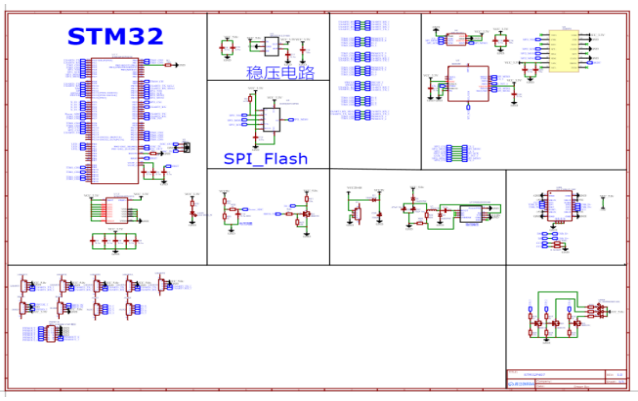
The UAV's flight control system is responsible for attitude control and navigation. It uses a PID adjustment algorithm to achieve precise control and is designed with dual-loop architecture supporting multi-channel data inputs. The schematic and PCB layout of the flight control board are shown below. The schematic uses the STM32F407VGT6 as the main controller, the BMI088 as the gyroscopic sensor for angle acquisition, the SPL06 as the altimeter for flight altitude control, and the AK8975 magnetometer for gyroscope calibration. Flight data is saved to an onboard Flash chip. Additionally, the flight control board includes a power voltage step-down circuit to reduce the battery voltage to the required level for the board. The flight control system also supports a one-button take-off function. When initiating take-off, the program first starts the motors, then gradually increases the electronic speed, and finally executes an altitude-holding program to allow the UAV to achieve a stable hover at a designated altitude.

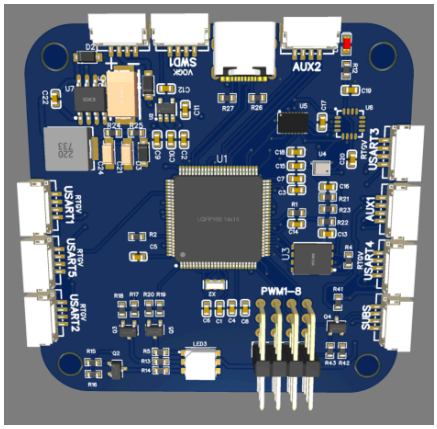
Advantages:

- **STM32 Microcontroller**: Compared to other similar products like Arduino or AVR controllers, the STM32 microcontroller offers higher computational power and more

hardware peripheral interfaces. It can handle more complex tasks, making it ideal for developing high-performance embedded applications.

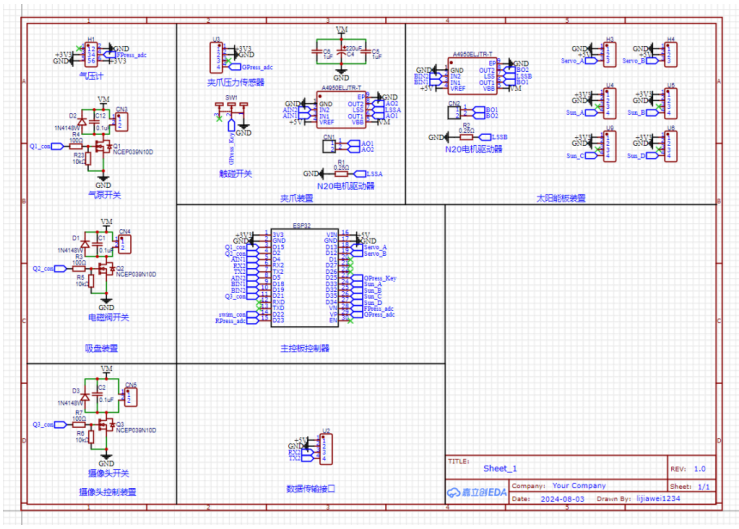
- **SPI Flash Storage**: Unlike traditional serial EEPROM, SPI Flash provides faster read/write speeds and larger capacities while consuming less power. This makes it especially suitable for systems requiring high-speed data storage and retrieval, such as firmware updates and data logging.

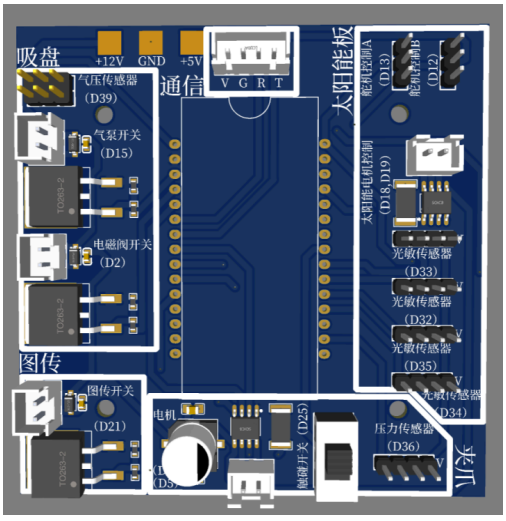




3.2 Circuit Design for Other UAV Modules

The UAV in this project integrates the control systems for the solar panel, suction cup device, claw mechanism, and flotation device into a single board. The main controller uses the ESP32 microcontroller and controls modules such as the air pump, solenoid valves, and video transmission camera through MOSFETs. This system suits embedded applications requiring wireless communication and multi-channel control, such as UAV control, automation devices, and intelligent control systems.





Sensor Components List: air pump, solenoid valve, image transmission camera, pressure sensor, motors/servos, ESP32, photosensitive sensor.

Advantages:

- **ESP32 Microcontroller**: Compared to traditional microcontrollers (e.g., Arduino or STM32), the ESP32 has unique advantages in wireless communication, with integrated Wi-Fi and Bluetooth functionality. It is particularly suitable for wireless control and data transmission applications, efficiently handling multi-tasking, and connecting multiple external modules.
- **MOSFET Driving Circuit**: Compared to relays, MOSFETs provide faster switching speeds, more precise control, and smaller sizes. They are ideal for applications with space constraints and high-frequency switching requirements. MOSFETs enable low-power, stable control of high-power loads, making them highly suitable for automation systems.

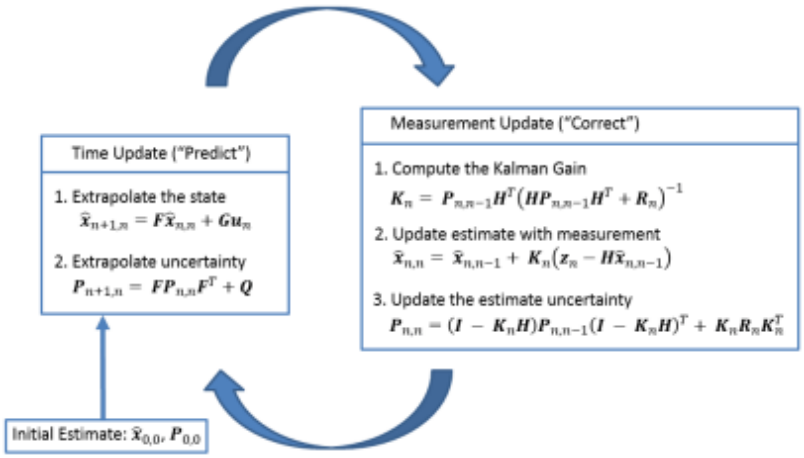
3.3 UAV Flight Control Algorithms

Quadcopter UAVs rely on multiple sensors to gather environmental information and control attitude. The accelerometer and gyroscope are the most common sensors, providing gravity acceleration and angular velocity data, respectively, to estimate roll, pitch, and yaw.

- The accelerometer measures Earth's gravitational acceleration to estimate pitch and roll angles.
- The gyroscope measures angular velocity to estimate yaw angle and compensates for long-term accelerometer errors.

Sensor fusion algorithms are commonly used to achieve accurate and stable attitude estimation. The two most common are:

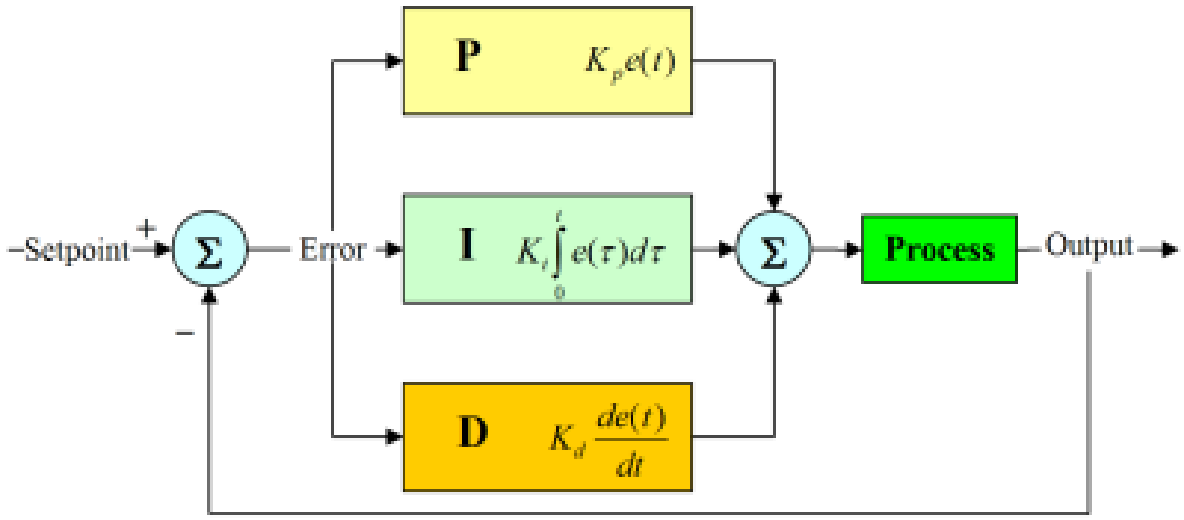
1. **Complementary Filter:** A simplified filtering algorithm that combines accelerometer and gyroscope data. It provides real-time, computationally efficient attitude estimation while maintaining stability.
2. **Kalman Filter:** A more complex algorithm that builds a mathematical system model, combining sensor data with noise information to optimize angle estimation, reduce errors, and achieve highly accurate attitude estimation.



The PID Controller (Proportional-Integral-Derivative Controller) is one of the most commonly used algorithms in flight control. It calculates the error between the target value (e.g., desired attitude) and the actual value (sensor-measured attitude) to generate control signals that adjust motor speed and maintain stability.

PID Controller Components:

- **Kp (Proportional Constant):** Affects system response speed.
- **Ki (Integral Constant):** Corrects long-term deviations to ensure stability.
- **Kd (Derivative Constant):** Suppresses oscillations and improves stability.



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4. Energy Management

4.1 Energy Management Strategy

The energy management system (EMS) is key to achieving "infinite endurance" for UAVs in the research platform, particularly in ensuring continuous energy supply and management. EMS guarantees a stable and efficient power supply during operations, significantly improving energy utilization and system reliability. The main functions of EMS include solar energy harvesting, conversion, and storage.

4.1.1 Hardware

The photovoltaic (PV) panel serves as the energy harvesting unit in EMS, primarily converting solar energy into electrical energy. Modern PV panels are typically made of high-efficiency monocrystalline or polycrystalline silicon materials, with a photoelectric conversion efficiency ranging between 20% and 25% under ideal conditions. The actual efficiency is influenced by

factors such as sunlight intensity, temperature, and the cleanliness of the PV panel surface. To maximize the photoelectric conversion efficiency, the design of the PV panel must consider factors such as installation angle, orientation, and adaptability to various climatic conditions.

The charging control and the battery protection module are responsible for monitoring and managing the battery charging process. Its primary function is to prevent overcharging and over-discharging, ensuring battery longevity and safety while avoiding damage due to improper charging. This module typically includes overcharge protection and current limiting, ensuring an efficient and safe charging process.

4.1.2 Energy Management Concept

In the energy management system, the sun-tracking algorithm is the core technology for optimizing PV panel efficiency. This algorithm controls motors and sensors to dynamically track the sun in real time, ensuring that the PV panel is always oriented towards the sun for maximum energy absorption. The sun-tracking algorithm typically employs Maximum Power Point Tracking (MPPT) technology, which continuously adjusts the PV panel's angle to adapt to changes in sunlight intensity. To optimize the utilization of harvested solar energy, EMS must implement an intelligent scheduling system to manage power distribution efficiently. EMS can monitor battery levels in real time by integrating with the flight control system and automatically balance power output with flight modes. For instance, the system can prioritize solar power usage under optimal sunlight, whereas it can switch to battery power in low-light conditions. This intelligent scheduling approach improves energy efficiency and extends the UAV's endurance.

4.1.3 Energy Flow Concept

The core objective of the EMS system is to ensure efficient solar energy harvesting, stable storage, and strategic utilization. EMS enables self-sustaining energy management through the coordinated operation of PV panels, charging control modules, and the sun-tracking algorithm. This allows UAVs to continuously receive sufficient power during prolonged missions, achieving "unlimited endurance."

Implementation process:

1. Solar energy is first harvested and converted into electrical energy by the solar panel.
2. The charging control module manages the electrical energy to ensure a safe and efficient charging process.
3. The battery management module monitors the stored energy in the battery.
4. The battery supplies power to the UAV system to support task execution.
5. The UAV's energy demands during operation trigger the sun-tracking algorithm, which is executed by electronic control hardware in real time.
6. The sun-tracking algorithm adjusts the angle of the PV panel to maximize solar energy absorption, forming a closed-loop control system.

4.2 Solar Tracking Algorithm

4.2.1 Introduction and Assumptions

This section details the methodology for optimizing the solar tracking system to maximize the energy harvested by the photovoltaic (PV) panels while minimizing energy losses due to servo adjustments. Our approach relies on full solar–position computations, avoiding simplified approximations. The following assumptions are made:

- **Ideal, Clear-Sky Conditions:** No atmospheric attenuation, weather effects, or shading are considered.
- **Constant PV Efficiency:** The PV panels operate at a constant efficiency throughout the day.
- **Instantaneous Servo Response:** Adjustments are assumed to occur immediately when the misalignment threshold is exceeded.
- **Fixed Variable Energy Cost for Adjustments:** Only the energy cost that varies with adjustment frequency (i.e., the energy required to overcome the system's inertia) is considered for optimization. Other constant energy costs (such as continuous servo holding energy) are not included since they do not affect the threshold optimization.

4.2.2 Solar Position Algorithm (SPA)[22]

At any time t, the sun's position is characterized by its elevation, $\alpha(t)$, and azimuth, $A(t)$. These are computed using full astronomical formulas:

Solar Elevation:

$$\sin\alpha(t) = \sin\phi \cdot \sin\delta(t) + \cos\phi \cdot \cos\delta(t) \cdot \cos H(t)$$

Where:

- ϕ is the observer's latitude,
- $\delta(t)$ is the solar declination (dependent on the day-of-year and time), and
- $H(t)$ is the solar hour angle (zero at solar noon).

Solar Azimuth:

$$\sin A(t) = \frac{\cos\delta(t) \cdot \sin H(t)}{\cos\alpha(t)} \quad \text{and} \quad \cos A(t) = \frac{\sin\delta(t) - \sin\phi \cdot \sin\alpha(t)}{\cos\phi \cdot \cos\alpha(t)}$$

(with appropriate quadrant corrections applied to determine the actual azimuth).

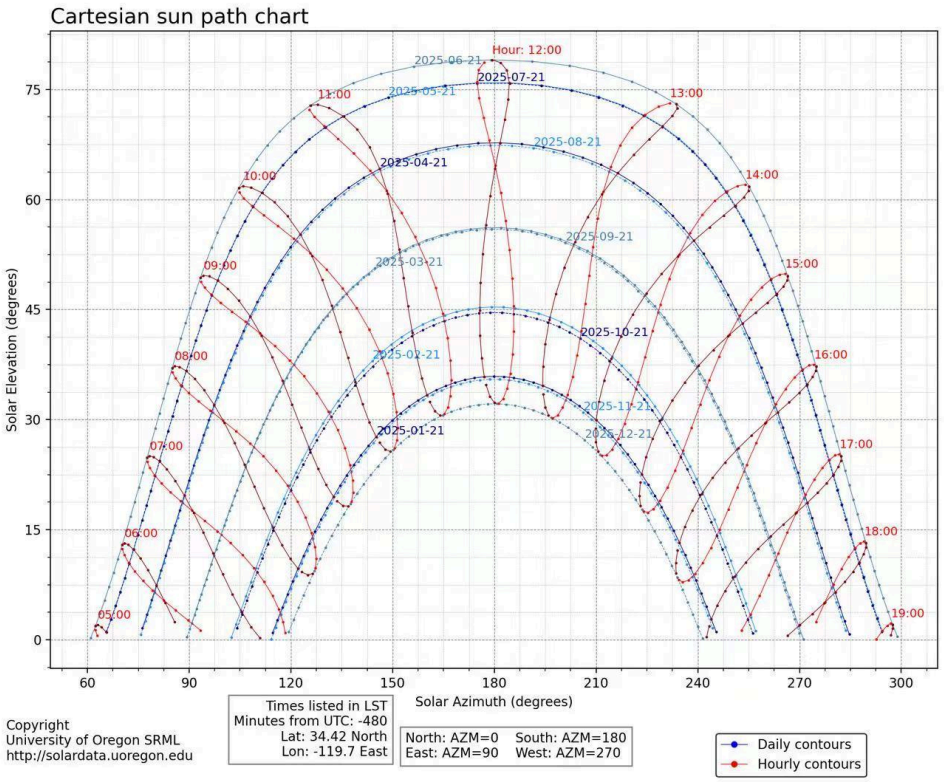
The sun's direction is then represented as the unit vector:

$$\mathbf{s}(t) = \Big(\cos\alpha(t), \sin A(t), ; \cos\alpha(t), \cos A(t), ; \sin\alpha(t)\Big).$$

This formulation captures the full, non–uniform motion of the sun.

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[22] Reda, I., & Andreas, A. (2004). Solar position algorithm for solar radiation applications. Solar Energy, 76(5), 577–589. https://doi.org/10.1016/j.solener.2003.12.003

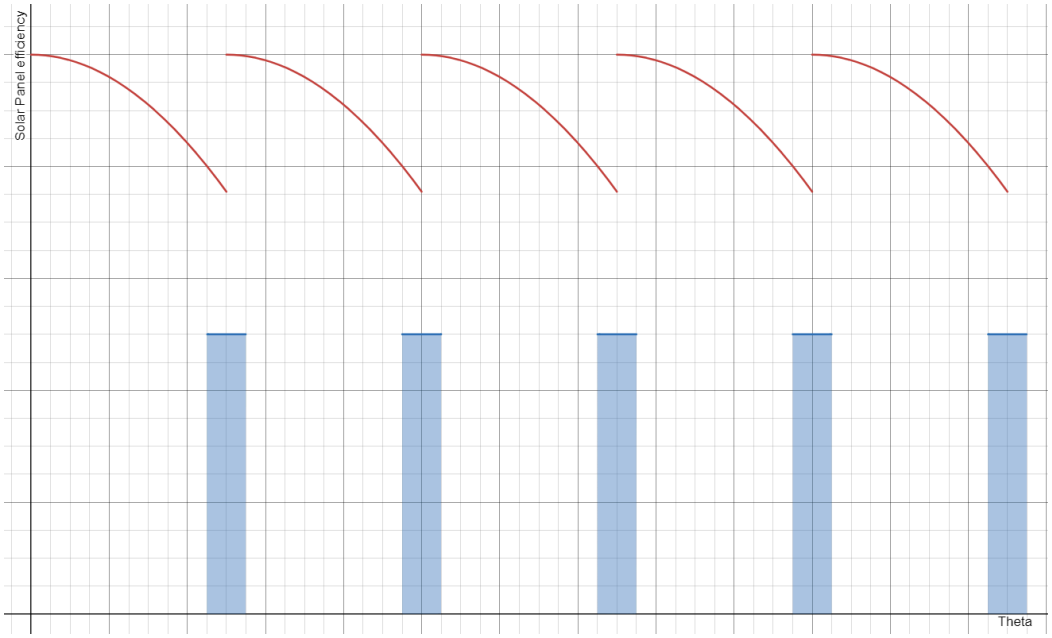


[23]

4.2.3 Tracking Cycles and Threshold Triggering

- **Initial Alignment:**
  At the start of each tracking cycle (time t_i), the PV panel is aligned so that its surface normal $\mathbf{n}(t_i)$ exactly matches the sun's direction $\mathbf{s}(t_i)$.
- **Monitoring Misalignment:**
  The sun's position changes as the day progresses while the panel remains fixed. The instantaneous incidence (misalignment) angle is computed as:
  $\theta_{\text{inc}}(t) = \arccos\Big(\mathbf{s}(t) \cdot \mathbf{n}(t_i)\Big)$.
- **Threshold Trigger:**
  A predefined threshold angle, $\theta_{\max}$, is set such that when $\theta_{\text{inc}}(t) > \theta_{\max}$, a servo adjustment is triggered. At that moment, a new tracking cycle begins with the panel reoriented to the current sun vector, i.e.,
  $\mathbf{n}(t_{i+1}) = \mathbf{s}(t_{i+1})$.

[23] University of Oregon Solar Radiation Monitoring Laboratory. (n.d.). UO SRML: Sun path chart. Retrieved March 5, 2025, from https://solardata.uoregon.edu/SunPathChart.html



The illustration demonstrates the cycles, and the red graph describes solar panel efficiency, while the blue bars represent the energy consumed when the angle is adjusted based on the threshold.

4.2.4 Energy Gained by the PV Panels

During each tracking cycle (from t_i to t_{i+1}), the power produced by the PV panels at given time is given by:

$$P(t) = P_{\max}, \cos\left[\theta_{\text{inc}}(t)\right],$$

Where $P_{\max}$ is the maximum power output under perfect alignment. The energy harvested in a single cycle is:

$$E_i = \int_{t_i}^{t_{i+1}} P_{\max}, \cos\left[\theta_{\text{inc}}(t)\right], dt.$$

The total energy gain over the day is

the sum of the energies from all cycles:

$$E_{\text{gain}}(\theta_{\max}) = \sum_{i=0}^{N-1} E_i.$$

4.2.5 Energy Lost Due to Servo Adjustments

To determine the energy cost of each adjustment, we first compute the effective moment of inertia, I_{eff}, of the rotating assembly. This comprises two components:

While a servo adjustment involves several energy components (e.g., energy for rotation and energy to hold the panel in position), only the variable cost—the energy used to overcome inertia—is considered for optimization because:

Variable Energy Cost: This cost depends on the number of adjustments and directly impacts net energy. More frequent adjustments lead to higher cumulative inertia-related energy loss.

Constant Energy Costs: Other energy expenditures (e.g., continuous servo operation during rotation or holding energy) remain constant regardless of the adjustment frequency. They cancel out in the optimization process and are therefore excluded from the net energy calculation.

Azimuth Component (I_{az}):
Assume the UAV's body is modeled as a rectangular prism with mass $m_{\text{az}} = 3.5, \text{kg}$ and effective dimensions $L_y = 0.2, \text{m}$ and $L_z = 0.2, \text{m}$, as most of the body and mechanisms are within this range. Its moment of inertia about an axis through its center is:
$I_{\text{az}} = \frac{1}{12}, m_{\text{az}}, \left(L_y^2 + L_z^2\right).$

Substituting the values: $I_{\text{az}} \approx \frac{1}{12} \times 3.5 \times \left(0.2^2 + 0.2^2\right) \approx 0.02333, \text{kgm}^2.$

Elevation Component (I_{el}):
Each solar panel is treated as a thin rectangular plate with mass $m_{\text{panel}} = 0.1, \text{kg}$ and a characteristic dimension $a = 0.2, \text{m}$ (about which it rotates). Its moment of inertia is:
$I_{\text{el}} = \frac{1}{12}, m_{\text{panel}}, a^2.$

Therefore: $I_{\text{el}} \approx \frac{1}{12} \times 0.1 \times (0.2)^2 \approx 0.000333, \text{kgm}^2.$

For two panels, the combined inertia is $I_{\text{el,total}} = 2 \times I_{\text{el}} \approx 0.000666, \text{kgm}^2.$

Thus, the effective moment of inertia is:
$I_{\text{eff}} = I_{\text{az}} + I_{\text{el,total}} \approx 0.02333 + 0.000666 \approx 0.024, \text{kgm}^2.$

Energy per Adjustment

The rotational kinetic energy gives the energy required for each servo adjustment (to overcome inertia): $E_{\text{adjust}} = \frac{1}{2}, I_{\text{eff}}, w^2,$

Where w is the angular velocity of the servo. For example, if the servo rotates 60° (1.0472 rad) in 0.098 s (yielding $w \approx 10.68, \text{rad/s}$), then: $E_{\text{adjust}} \approx \frac{1}{2} \times 0.024 \times (10.68)^2 \approx 1.37, \text{J}.$

Thus, the total energy loss due to adjustments over the day is:
$E_{\text{servo}}(\theta_{\max}) = N(\theta_{\max}) \times E_{\text{adjust}},$
Where $N(\theta_{\max})$ is the number of tracking cycles (i.e., adjustments).

4.2.6 Net Energy Formulation and Conclusion

The net energy delivered by the solar tracking system is the difference between the total energy gained by the PV panels and the cumulative energy lost due to servo adjustments:
$E_{\text{net}}(\theta_{\max}) = E_{\text{gain}}(\theta_{\max}) - E_{\text{servo}}(\theta_{\max}).$
In expanded form, this is:
$$E_{\text{net}}(\theta_{\max}) = \sum_{i=0}^{N-1} \left(\int_{t_i}^{t_{i+1}} P_{\max}, \cos\left[\theta_{\text{inc}}(t)\right], dt \right) - N(\theta_{\max}) \times E_{\text{adjust}}.$$

The optimal tracking threshold, $\theta_{\max}$, is the value that maximizes E_{net} by balancing the increased energy capture from more frequent adjustments against the extra energy loss incurred per adjustment. This comprehensive formulation underpins our threshold-triggered dual-axis solar tracking system.

4.3 Static Modules in Azimuth Axis

4.3.1 Claw

In the claw state, the UAV hangs vertically from a horizontal branch, maintaining a stable and controlled suspension. The azimuth servo enables full-range adjustment of the UAV's orientation, allowing precise tracking of targets by adjusting both azimuth and elevation angles. This precise control ensures stable operation, even in confined spaces like tree branches, enhancing observational and operational accuracy.

4.3.2 Suction Cup

The UAV relies on the azimuth servo for directional adjustments when attached to a vertical surface. While the servo theoretically allows 360-degree rotation, wall constraints may limit movement. An electronic fence mechanism monitors and restricts the azimuth angle to mitigate collision risks, dynamically adjusting elevation as needed. This system ensures safe and stable operation in constrained environments.

4.3.3 Floating/Buoyancy

On the water, the UAV remains buoyant using an inflatable ring. The azimuth servo does not play a role in adjusting the UAV's heading while floating. The air pump and propulsion system enable precise directional control, ensuring stability and maneuverability for aquatic tasks like surveying and rescue operations.

4.4 Other Energy Conserved

4.4.1 Suction Cup

The suction cups are airtight, and the pump is only used periodically when the interior pressure level rises above the target pressure.

4.4.2 Claw

The claw utilizes a worm gear mechanism and can be powered off when not actuating the claw.

4.4.3 Buoyancy Device

The device is kept airtight and will only need to be reinflated periodically.

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5. Experiments and Results

5.1 Stability Verification

This experiment aims to evaluate the stability of a UAV under different flight conditions. The experiment analyzes attitude variations and stability indicators by conducting static and dynamic flight tests, providing data support for optimizing the UAV's control system.

Experimental Steps:

1. **Equipment Inspection:** Conduct a thorough inspection of the UAV and its control system to ensure proper hardware and software functionality for safe operation.
2. **Static Testing:** Perform hover tests in an open testing area and record the UAV's attitude stability and attitude errors while stationary.
3. **Dynamic Testing:** Execute a series of flight maneuvers, including straight-line movement, turns, and sudden stops, to measure the UAV's attitude adjustment capability and flight stability under different motion states.



Open Indoors Testing



Narrow Indoors Testing

This experiment helps analyze the UAV's adaptability to complex flight environments and provides experimental evidence for further flight control algorithm improvements.

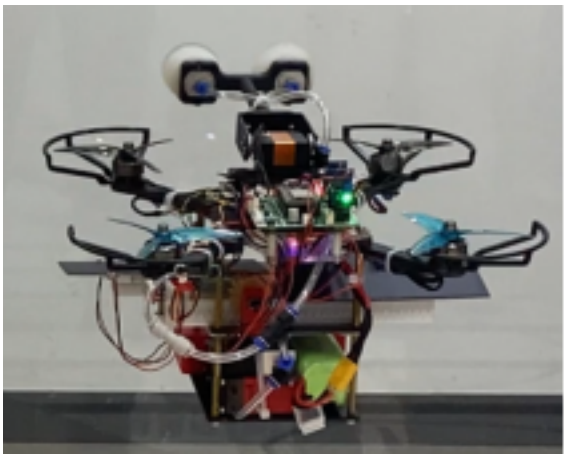
5.2 Static Experiments

5.2.1 Suction Cup Experiment

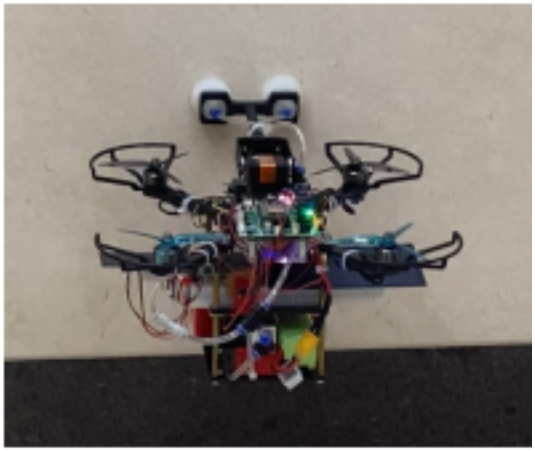
This experiment aims to verify the suction cup capability, response efficiency, and energy consumption of the detachable suction cup installed on the UAV when attached to two different surfaces: tile and glass. The suction cup module uses a 13L/min air pump and a pressure feedback system. Energy consumption and response time are recorded during the adsorption and detachment process.

Experimental Steps:

1. **Check** the integrity and functionality of the UAV and suction cup.
2. **Select** two test materials: tile and glass.
3. **Install** the suction cup on the top of the UAV and ensure a secure fit.
4. **Conduct** tests on both tile and glass surfaces, performing three tests for each material.
5. **Record** the success rate of transitioning from flight mode to static state.
6. **Measure and compare** the suction time for different materials.
7. **Analyze the data** to evaluate the suction cup's performance on different surfaces.



Glass



Tile

Table 1: Suction Cup Tests

Adsorption material	Adsorption state	Possible to achieve the flying to static action	Suction time
Tile	Pure adsorption state	Y	1m25s

Tile	Pressure control status	Y	>30min
Tile	Pure static state	Y	1m12s
Tile	Pressure control status	Y	>30min
Tile	Pure static state	Y	1m34s
Tile	Pressure control status	Y	>30min
Glass	Pure static state	Y	5m11s'
Glass	Pressure control status	Y	>30min
Glass	Pure static state	Y	4m43s
Glass	Pressure control status	Y	>30min
Glass	Pure static state	Y	4m57s
Glass	Pressure control status	Y	>30min

From Table 1, it can be observed that the adsorption time on tiles was 1m25s, 1m12s, and 1m34s, whereas on glass, the adsorption time was significantly longer at 5m11s, 4m43s, and 4m57s. In the controlled suction cup state, the adsorption time exceeded 30 minutes for both materials, demonstrating the suction cup's strong performance across different surfaces. Notably, the suction cup exhibited better stability on glass surfaces. Total power consumption: 119mAh.

5.2.2 Claw Experiment

This experiment aims to test whether the UAV's claw can successfully grasp PVC pipes of different sizes, evaluating its gripping performance and stability. The experiment involves inspecting the UAV and the claw, preparing PVC pipes of 110mm, 75mm, and 80mm diameters, and installing the gripper for grasping tests.

Experimental Steps:

1. **Check** the integrity and functionality of the UAV and claw.
2. **Prepare** PVC pipes of three sizes: 110mm, 75mm, and 80mm.
3. **Install** the claw onto the UAV.
4. **Conduct** grasping tests for each pipe size.
5. **Record** the success rate of the gripping process.



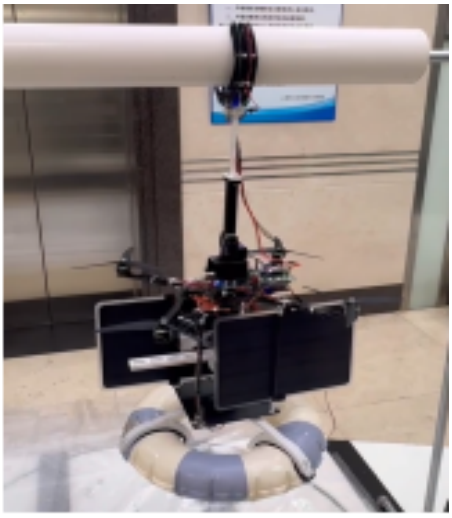


Table 2: Claw Tests

PVC pipe size	Whether the trial was successful or not
110mm	S
80mm	S
75mm	S
110mm	S
80mm	S
75mm	S
110mm	S
80mm	S
75mm	S

The gripper demonstrated strong gripping force across different PVC pipe sizes, achieving a high success rate. This confirms the gripper's stability and effective self-locking capability, particularly when handling larger-diameter pipes. The gripper maintains sufficient grip strength to ensure reliable operation—total power consumption: 52mAh.

5.3 Solar Tracking Optimization

5.3.1 Simulation Setup

To validate the solar tracking model and determine the optimal tracking threshold, a numerical simulation was conducted with the following parameters:

- **Date:** March 7, 2025
- **Location:** Santa Barbara, California (Latitude: 34.42° N, Longitude: -119.70° W)
- **PV Panel Power:** $P_{\max} = 20, \mathrm{W}$ (combined output of two 10-W panels).
- **Servo Adjustment Energy:** $E_{\mathrm{adjust}} \approx 1.37, \mathrm{J}$ (energy cost per adjustment, computed from the effective moment of inertia and angular velocity).
- **Time Resolution:** A time step of 10 seconds was used.
- **Time Window:** The simulation was run from local sunrise at 6:19 AM to sunset at 6:00 PM.
- **Method:**
    1. The solar position (elevation and azimuth) was computed at each time step using the full solar position algorithm (SPA).
    2. At the beginning of the simulation, the panel is aligned with the sun. As time progresses, the instantaneous misalignment angle
    $$\theta_{\mathrm{inc}}(t) = \arccos\Big(\mathbf{s}(t) \cdot \mathbf{n}(t_i)\Big)$$
    It is calculated at each time step.
    3. When the misalignment exceeds a candidate tracking threshold, $\theta_{\max}$, a servo adjustment is triggered (incurring an energy cost of E_{adjust}), and the panel's orientation is reset to the current sun vector.
    4. The energy produced during each cycle is integrated, and the total net energy is computed as:
    $$E_{\mathrm{net}}(\theta_{\max}) = E_{\mathrm{gain}}(\theta_{\max}) - N(\theta_{\max}) \times E_{\mathrm{adjust}},$$
    Where $N(\theta_{\max})$ is the number of adjustments.

5.3.2 Simulation Results

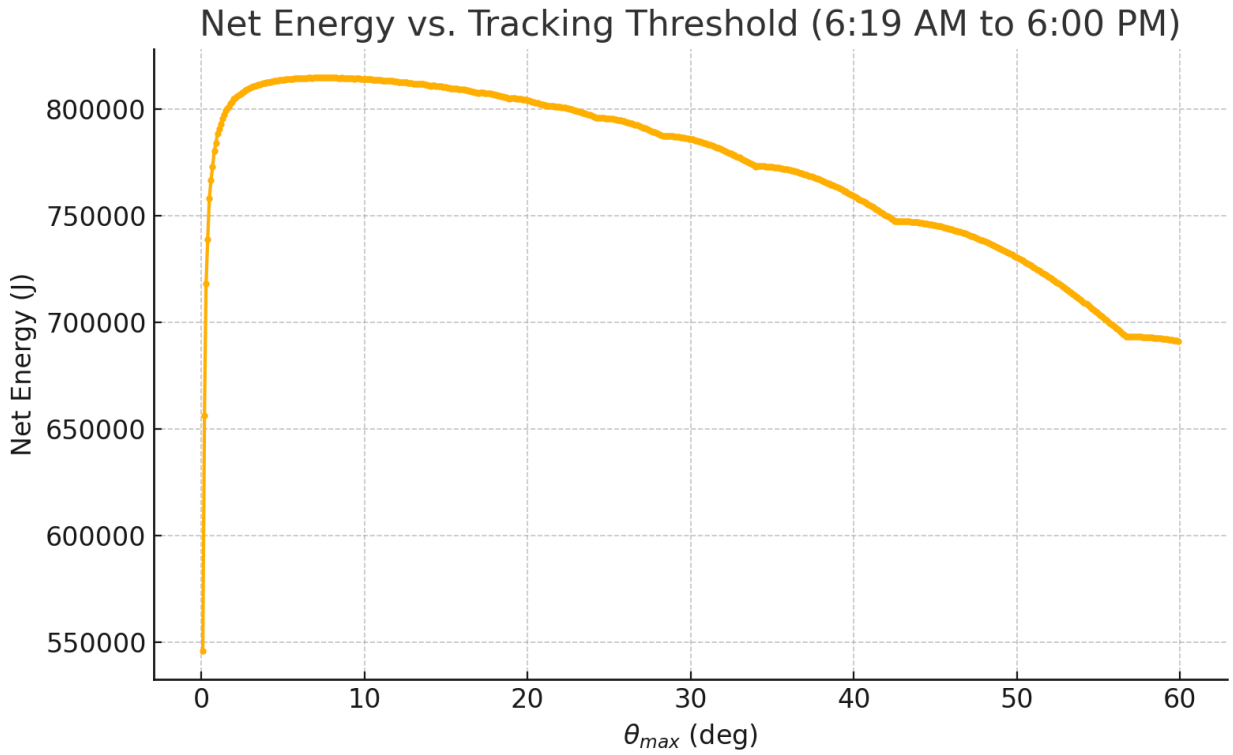
The simulation swept candidate thresholds from 0.1° to 60° to determine the value that maximizes net energy. The results indicated that the optimal tracking threshold is approximately:

$$\theta_{\max} \approx 7.40^{\circ}$$

At this threshold, the total net energy captured over the day was found to be approximately:

$$E_{\mathrm{net}} \approx 814703.52, \mathrm{J}$$

The following plot illustrates the net energy as a function of the tracking threshold:



5.3.3 Analysis

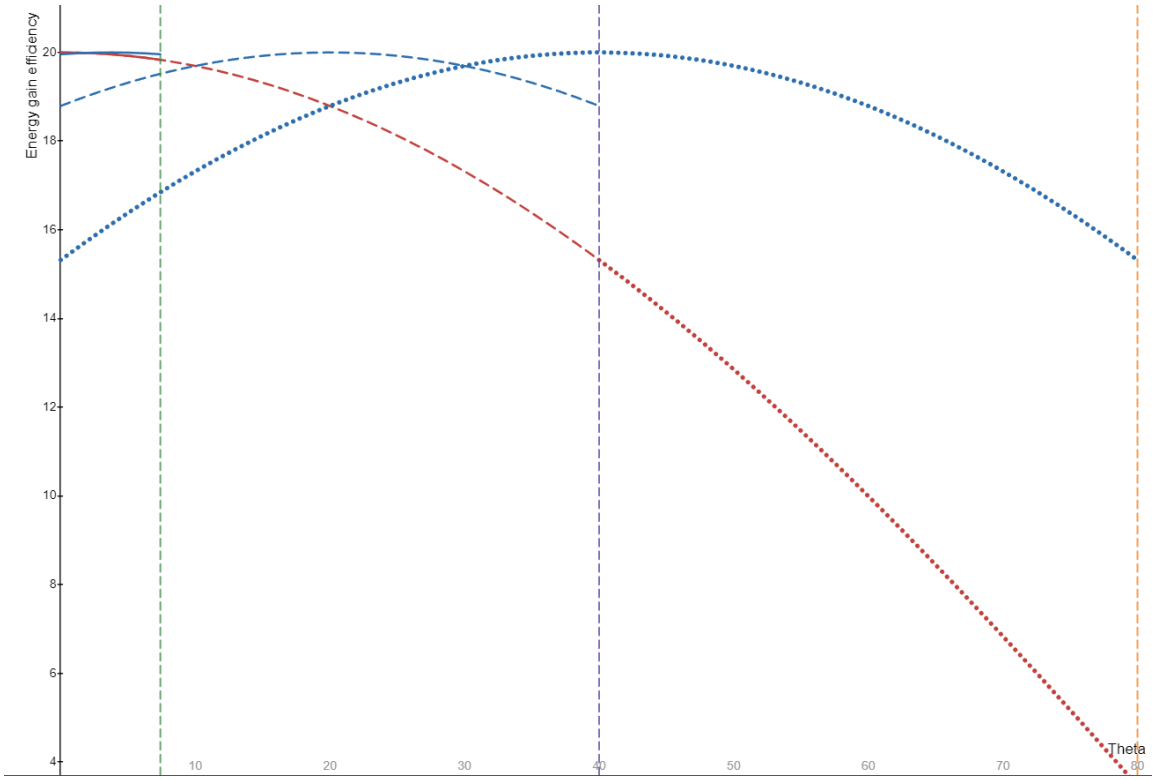
- **Threshold Effect:**
  At very low thresholds, frequent adjustments occur, resulting in high cumulative energy losses due to the inertia-related cost (E_{adjust}). In contrast, very high thresholds allow significant misalignment, thereby reducing the effective power capture from the PV panels.
- **Optimal Value:**
  The simulation shows that a threshold of approximately 7.40° provides the best balance between energy capture and servo energy loss, maximizing the net energy.
- **Robustness:**
  Under the ideal clear-sky conditions assumed in the simulation, the full solar position data validates the model's accuracy in accounting for the sun's non-uniform motion.

Conclusion:
The experimental results demonstrate that the UAV's PV system can maximize net energy output by adopting a threshold-triggered dual-axis solar tracking strategy and optimizing the tracking threshold to about 7.40°. This optimization is crucial for extending the UAV's endurance by balancing improved energy capture against energy losses from servo adjustments.

5.3.4 Exploration in Other Tracking Techniques

Another approach towards solar tracking cycles with an angle threshold trigger involves pre-aligning the panels to an offset of $\frac{\theta_{\max}}{2}$ ahead of the current sun position. This means that

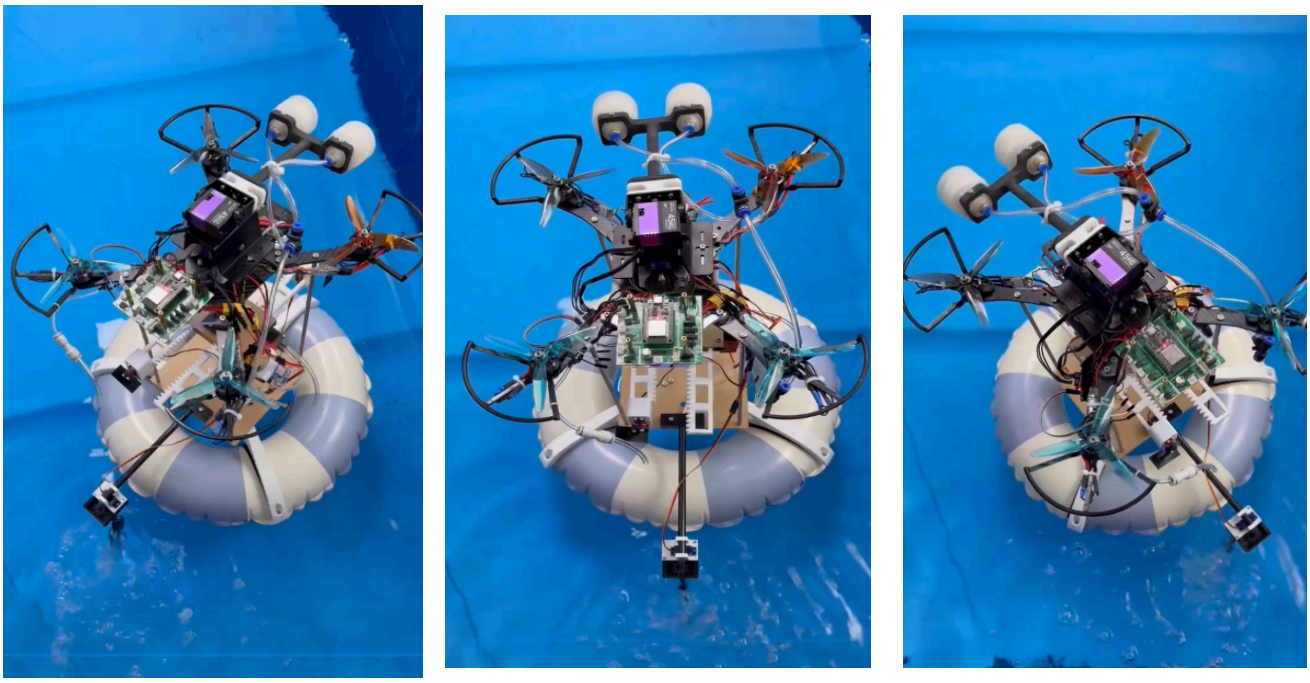
the solar panels will readjust to $\frac{\theta_{\max}}{2}$ ahead of the sun when the θ_{inc} is equal to $\frac{\theta_{\max}}{2}$. In this scenario, the adjusted angle per cycle is still $\theta_{\max}$, not changing the total adjustments in a given day; thus, the E_{adjust} should remain the same no matter the method used. When viewed graphically, the new method's instantaneos panel energy efficiency takes the flattest part of the cosine curve (the peak), while the current method just takes the cosine curve from 0 to $\theta_{\max}$. In the current method, efficiency is highest only at the precise moment of realignment, followed by a gradual decline until the next correction. In contrast, the new method extends periods of high efficiency on both sides of the peak, ensuring more consistent energy absorption. If the integral is taken and E_{gain} is found in relationship to $\theta_{\max}$ for both methods, the new method will always gain more energy and will gain significantly more as $\theta_{\max}$ increases.



Green (Solid Line, $\theta_{\max}$=7.4): The optimal angle determined by simulation. This configuration provides a **0.2% increase** in energy efficiency, a minor but measurable improvement.
Purple (Dashed Line, $\theta_{\max}$=40): A moderately sized adjustment, yielding a **6.4% increase** in energy capture.
Orange (Dotted Line, $\theta_{\max}$=80): A large misalignment, resulting in a **30.5% energy gain**.

Implementation & Plans
Although this alternative tracking method is not yet implemented, initial findings suggest that while it increases algorithmic complexity, it can provide noticeable energy gains. The trade-off is minimal at smaller angles but becomes increasingly beneficial when the threshold angle increases, making it worthwhile for integration. Given its demonstrated efficiency, the tracking system will incorporate this method when the predicted gains justify the added complexity.

5.4 Buoyancy Experiments



The UAV can float and propel itself through the pump system.

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6. Discussion, Analysis, and Limitations

6.1 Limitations

6.1.1 Overall Design

The UAV’s design is inherently tall, which may lead to balancing issues, especially in specific conditions (e.g., during flight maneuvers, changes in altitude, or when interacting with environmental factors). The increased height could also limit the UAV's maneuverability in confined or complex spaces, reducing its overall operational flexibility.

6.1.2 Suction Cup

The suction cup's ability to adhere to surfaces depends on the surface's material and smoothness. It struggles with more porous or rougher surfaces, leading to unreliable attachment unless constant pumping is maintained. This continuous pumping consumes significant energy, which could drain the battery quickly in extended operations. Moreover, using the azimuth axis for solar adjustments becomes challenging as the suction cup setup may hinder the free movement or rotation of the azimuth axis.

6.1.3 Claw
The claw uses a worm gear mechanism, which is reliable in holding strength but relatively slow when actuating. This slower actuation can be problematic in time-sensitive tasks requiring faster response times. Optimization of the actuation speed might be needed for more dynamic operations.

6.1.4 Flotation Device
The UAV's buoyancy system allows it to operate on water but faces stability challenges in rough water conditions. Waves or instability in the water's surface could lead to dangerous or unpredictable UAV behavior. In addition, the buoyancy device has limited ability to adjust to highly dynamic water surfaces, reducing its effectiveness for certain tasks. Furthermore, the buoyancy system introduces additional constraints for solar tracking—accurate tracking becomes difficult when the UAV is floating, as movements from water waves or surface instability prevent precise alignment with the sun.

6.1.5 Dual Camera Gimbal Mounts
The dual-camera gimbal system faces some practical limitations in terms of accessibility and functionality. The UAV itself can obstruct the camera's field of view, particularly when rotating the azimuth during solar tracking adjustments. In addition, the gimbal must account for these rotations, which can complicate maintaining stable and uninterrupted camera angles.

6.1.6 Solar Tracking System
As modeled in theoretical simulations, the solar tracking system assumes ideal conditions, including perfect weather. In reality, energy capture from the solar panels will likely be lower due to environmental factors, such as cloud cover, dust, or atmospheric interference. Non-sunny days will severely reduce the effectiveness of the solar tracking system. Furthermore, the energy captured from the panels is slower than that of other power sources. While the platform offers a stable and safe charging environment, it does not guarantee fast or consistent charging speeds under all conditions. The reliance on sunlight means that the UAV's endurance is highly weather-dependent.

6.2 Next Steps

6.2.1 Overall Design Optimization
To address the potential balancing issues and limitations in maneuverability, a redesign of the overall size and weight of the platform is needed. This would enhance stability and ensure more efficient operation in varying environments. More compact and balanced designs could improve the UAV's versatility, especially in constrained spaces, and improve performance in dynamic conditions.

6.2.2 Solar Panel and Tracking System Enhancement
To maximize energy gain and optimize solar panel usage, further research and development can be done to improve the tracking mechanism or algorithm. Exploring independent azimuth and elevation tracking would allow for more precise solar tracking and reduce energy loss from servo adjustments. Additionally, experimenting with alternative tracking techniques, such as

right ascension and declination tracking, could streamline the design by turning it into a one-axis system while maintaining similar efficiency. Though more complicated in structure, this could prove valuable for improving energy capture while minimizing energy consumption.

6.2.3 AI-Driven Image Recognition and Camera Guidance System

Integrating an AI-based image recognition and guiding system could enhance the dual-camera gimbal system. In this setup, the wide-angle video transmission camera would continuously scan the area for wildlife. Once a target is identified, the AI system recognizes the animal (e.g., a bird) and dynamically guides the main recording camera to either zoom in on the subject or follow its motion. The system would also account for the animal's distance and size, adjusting the recording camera's zoom and focus to ensure high-quality and stable footage. This AI-driven system could significantly improve the camera's real-time performance, providing more reliable and efficient wildlife observation and documentation.

6.2.4 Weatherproofing

Due to most of the time the UAV spent in the wild-life observation application occurring outside, the UAV will require weatherproofing to ensure its durability during operations.

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7. Conclusion

This study aims to design and implement a self-sustaining UAV platform powered by photovoltaic energy to meet the needs of long-term, efficient execution of scientific missions in outdoor environments, especially for bird monitoring tasks in the Santa Barbara area of California. Based on a conventional UAV system, the platform integrates multifunctional modules such as photovoltaic panels, suction cups, grippers, and buoyancy devices, endowing it with the ability to adapt to the environment and autonomously recharge, thereby effectively extending its endurance and improving the stability and reliability of mission execution.

To enhance energy utilization efficiency, the system employs a threshold-triggered dual-axis solar tracking strategy combined with the Maximum Power Point Tracking (MPPT) algorithm to optimize the process of harvesting photovoltaic energy, ensuring maximum energy conversion efficiency under varying lighting conditions.

The study first introduces innovations in structural design by using multimodal simulation to optimize the UAV’s mechanical structure, enabling it to adapt to different operational environments. Specifically, the platform incorporates multimodal structures such as a gripper system (for clinging and docking), a suction cup mechanism (for securing onto specific surfaces), and a buoyancy device (for stable hovering in aquatic environments), and these functionalities have been physically validated through experiments.

On the control system side, the study adopts the UAV’s flight control system as the core actuator. It utilizes an ESP32 microcontroller to communicate via serial communication with the flight control system to achieve precise control of the multimodal structures. Based on IO signal

operations, the system can flexibly adjust the flight mode and attachment strategies under different environmental conditions, enhancing the UAV's adaptability and mission execution capabilities in the field.

Extensive experiments were conducted to validate the system's performance, covering tests on photovoltaic energy harvesting efficiency, evaluating the UAV's autonomous docking and charging capabilities, and analyzing flight stability in different environments. The experimental results indicate that the platform can significantly improve the utilization of photovoltaic energy and achieve stable endurance in various complex settings, offering a feasible solution for long-term autonomous scientific missions in outdoor environments.

The innovations of this study include:

1. **Autonomous Endurance:** Enabling long-term endurance based on photovoltaic charging and a solar tracking system.
2. **Multimodal Structural Design:** Integration of grippers, suction cups, and buoyancy devices enhances the UAV's adaptability in various environments.
3. **Intelligent Control System:** Integrating the ESP32 with the flight control system to achieve precise control and adaptive adjustments of the multimodal structures.
4. **Energy Optimization Strategy:** Combining MPPT algorithms with a dual-axis solar tracking mechanism to improve photovoltaic energy conversion efficiency.

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